DATAWISE AI

DataWise AI: ML Methodology Report

Mathematical Foundations, Leakage Prevention, Feature Engineering & Optimization

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Orchestration:	LangGraph 16-Agent Autonomous Architecture

Architectural Foundation: Deterministic ML execution engine paired with autonomous multi-agent reasoning, strict data leakage boundaries, and human-in-the-loop validation.

1. Data Profiling & Statistical Intelligence

Prior to initiating transformation or training, DataWise AI conducts thorough exploratory data diagnostics via `DatasetProfilingAgent` and `DataQualityAgent`. These diagnostics detect missingness patterns, calculate skewness and kurtosis, compute mutual information with candidate targets, and identify severe multicollinearity using Variance Inflation Factor (VIF).

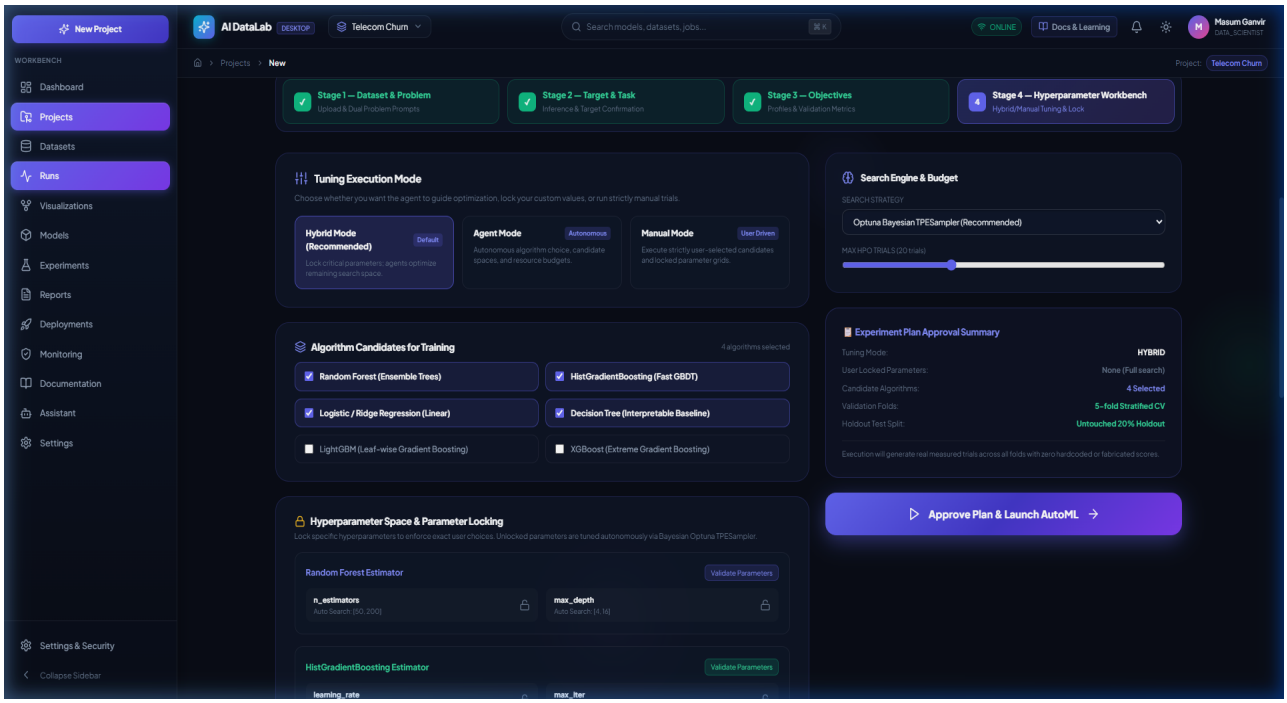


Figure: Machine Learning Diagnostics & Visual Exploratory Analysis

2. Preprocessing & Leak-Free Transformation Pipeline

A critical vulnerability in automated data science is data leakage. DataWise AI strictly separates splitting from fitting:

Pipeline Stage	Implementation	Leakage Prevention Mechanism
Pipeline Stage	Deterministic Implementation	Leakage Prevention Mechanism
Train/Val/Test Split	StratifiedShuffleSplit / TimeSeriesSplit	Split occurs before computing any statistics or parameters
Numeric Imputation	SimpleImputer (median) / KNNImputer	Imputation parameters computed strictly on X_train
Categorical Encoding	OneHotEncoder (handle_unknown='ignore')	Unknown test categories mapped to zero-vector without crashing
Feature Scaling	StandardScaler / RobustScaler	Mean and variance statistics fitted strictly on X_train
Target Encoding	TargetEncoder (with K-fold smoothing)	Uses inner cross-validation folds to prevent direct target memorization
Feature Selection	SelectFromModel / Mutual Information	Importance thresholds calculated strictly on training fold data

3. Supported Algorithmic Families

DataWise AI supports diverse algorithm families configured to balance accuracy, training throughput, and interpretability:

- **Linear & Logistic Models:** LogisticRegression, Ridge, Lasso, ElasticNet with L1/L2 regularization.
- **Tree Ensembles:** RandomForestClassifier/Regressor, ExtraTrees, GradientBoosting.
- **High-Performance Gradient Boosting:** XGBoost (with GPU acceleration support) and LightGBM.

4. Optuna Bayesian Hyperparameter Optimization

Hyperparameter tuning in DataWise AI is orchestrated by `ModelTuningAgent` using Optuna. The search optimizes objective functions over stratified K-fold cross-validation. Optuna's Tree-structured Parzen Estimator (TPE) algorithm models conditional probabilities of parameter choices, while MedianPruner halts poorly performing trials early.